

Instructions for *ACL Proceedings

Anonymous ACL submission

1 Introduction

This project tackles the task of classifying cultural items into three taxonomic classes (Cultural Agnostic (C.A), Cultural Representative (C.R), and Cultural Exclusive (C.E)). Our approach combines a large language model-based classifier and a knowledge-based, non-LM method exploiting structured data from Wikidata.

We aim to compare the strengths and limitations of each method, focusing on generalization, interpretability, and performance.

2 Methodology

2.1 LM-Based Classifier

We fine-tuned a pre-trained distilbert-base-cased model for the multiclass classification task using item-level metadata from Wikipedia. Our methodology included a preprocessing pipeline, input formatting with a Transformer tokenizer, and partial layer freezing to retain general semantic knowledge while specializing to the cultural classification task.

Preprocessing and Tokenization

Each data point included the item name, description, type, and category. We concatenated these fields into a single input string.

The concatenated strings were then tokenized using Huggingface's DistilBertTokenizer, with inputs truncated or padded to a maximum sequence length of 128 tokens. The tokenizer maps tokens to IDs, applies attention masks, and ensures that inputs are aligned for batch processing.

Model Architecture and Training Setup

To prevent catastrophic forgetting and reduce overfitting on a relatively small training set, we froze the initial 3 layers of the Transformer encoder, allowing only the top layers and the classification head to be fine-tuned. This ensures

that core linguistic representations are preserved while adapting the model to the downstream task. We trained using Huggingface's Trainer API, with cross-entropy loss, early stopping, and automatic logging. The model was evaluated on a stratified dev set using macro-averaged F1 and accuracy metrics.

DistilBERT vs RoBERTa

Additionally, we experimented with a second encoder-based model, roberta-base, following the same data processing and training configuration. While both models showed comparable learning behavior, the DistilBERT-based model outperformed RoBERTa on validation accuracy, and was therefore selected as the primary LM-based classifier for further evaluation and reporting.

2.2 Non LM-based Classifier

We employed multiple feature extraction techniques and machine learning models to perform a multiclass classification task using item-level metadata. The features include item-level text data, semantic information from Word2Vec embeddings, and additional metadata retrieved from Wikidata. We tested different models, including Logistic Regression, Random Forest, and Support Vector Machine (SVM), and selected the best-performing model based on validation accuracy.

Preprocessing and Feature Extraction

The input data included multiple columns: item name, description, type, and category. These fields were concatenated to create a single input string for each item. A text cleaning function was applied to remove stopwords from the concatenated text.

We extracted three types of features:

- **TF-IDF Features:** We used the TfidfVectorizer from scikit-learn to convert the preprocessed text data into numerical vectors. A

079	maximum of 5000 features was selected to	3 Experiments	126
080	prevent overfitting.		
081	• Word2Vec Embeddings: Word2Vec embed-	LM-based Classifier	127
082	dings were trained on the text data to capture	For the LM-based classifier, we fine-tuned both	128
083	semantic relationships between words. Each	DistilBERT and RoBERTa models for 3 epochs	129
084	text was transformed into a fixed-size vector	with a batch size of 64, weight decay set to 0.01	130
085	by averaging the embeddings of the words in	for regularization, and metrics logging every 10	131
086	the text.	steps. We used Huggingface’s Trainer API with	132
		early stopping enabled.	133
087	• Wikidata Features: Metadata from Wikidata	Evaluation was performed on both the training	134
088	was utilized to extract country-related prop-	and validation sets, using standard classification	135
089	erties (e.g., country of origin, citizenship) as-	metrics: accuracy, precision, recall, F1-score, and	136
090	sociated with the items. We used caching to	loss. In addition, we computed the confusion	137
091	avoid excessive API calls during data prepro-	matrix on the dev set to better understand the	138
092	cessing.	model’s prediction patterns across the three classes.	139
			140
093	These features were combined into a single input	Non LM-based Classifier	141
094	vector for each data point, resulting in a rich	For the non-LM pipeline, we observed a class	142
095	representation of each item, including both textual	imbalance in the training data, and therefore ap-	143
096	and metadata information.	plied SMOTE to oversample the minority classes	144
097		before training. The selected classifier (SVM) was	145
098	Model Architecture and Training Setup	then evaluated using the same metrics: accuracy,	146
099	We experimented with three different models:	precision, recall, F1-score, and confusion matrix	147
		analysis to gain deeper insights into class-wise	148
100	• Logistic Regression: A Logistic Regression	prediction behavior.	149
101	model with cross-validation was trained using		
102	the combined features, which included TF-	4 Results	150
103	IDF, Word2Vec embeddings, and Wikidata	4.1 LM-Based	151
104	metadata.	• Accuracy: 0.770	152
105	• Random Forest Classifier: A Random Forest	• Precision: 0.754	153
106	model with 100 estimators was also trained	• Macro F1: 0.752	154
107	on the combined features.	• Recall: 0.758	155
108	• Support Vector Machine (SVM): An SVM	• Best performance on Cultural Agnostic	156
109	classifier with a linear kernel was trained on	• Lower recall for Cultural Exclusive, due to its	157
110	the combined features.	sparsity and linguistic ambiguity.	158
111	For each model, the data was balanced using the	4.2 Non-LM Based	159
112	Synthetic Minority Over-sampling Technique	• Accuracy: 0.66	160
113	(SMOTE) to address class imbalance, and hyper-	• Macro Precision: 0.64	161
114	parameters were tuned to optimize performance.	• Macro F1: 0.64	162
115	The models were trained and evaluated using a	• Macro Recall: 0.64	163
116	stratified validation set.	• Best performance on Cultural Agnostic	164
117		• Lower recall for Cultural Exclusive, due to its	165
118	SVM vs Other Models	sparsity and linguistic ambiguity.	166
119	After evaluating all models, the Support Vector		
120	Machine (SVM) classifier performed the best		
121	on the validation set in terms of accuracy. The		
122	classification report for SVM showed the highest		
123	accuracy and F1-score compared to Logistic		
124	Regression and Random Forest, leading us to		
125	select SVM as the final model for deployment.		

A Appendix

A.1 Dataset Classes Distribution

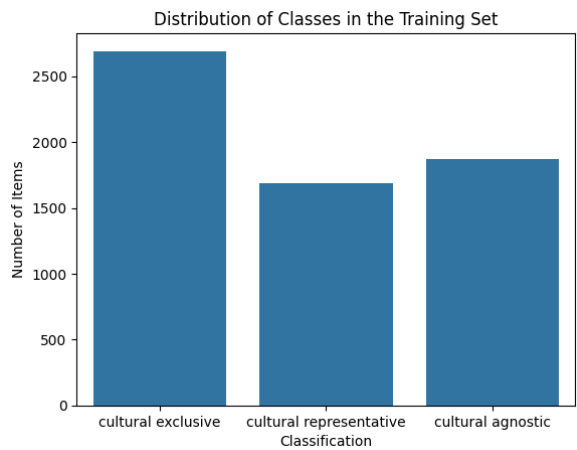


Figure 1: Dataset Classes Distribution

A.2 Classification Metrics

A.2.1 LM-Based Model

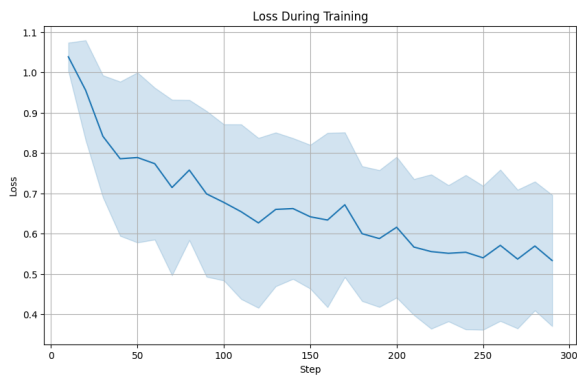


Figure 2: Training Loss

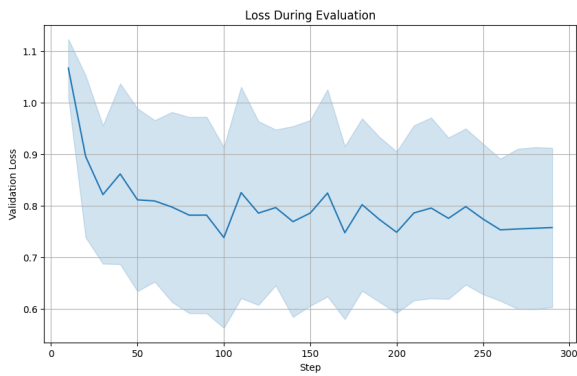


Figure 3: Validation Loss

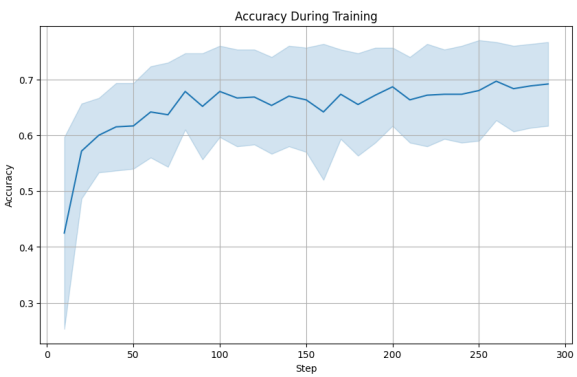


Figure 4: Accuracy

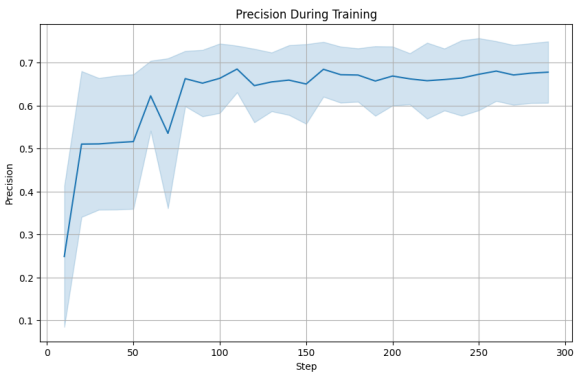


Figure 5: Precision

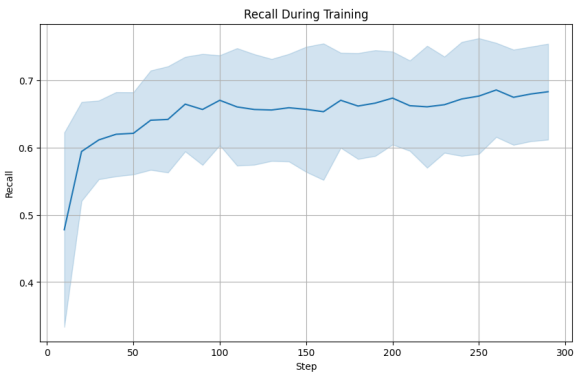


Figure 6: Recall

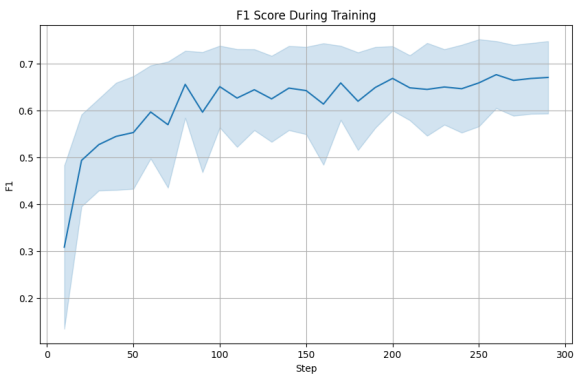


Figure 7: F1 Score

A.2.2 Non LM-Based Model

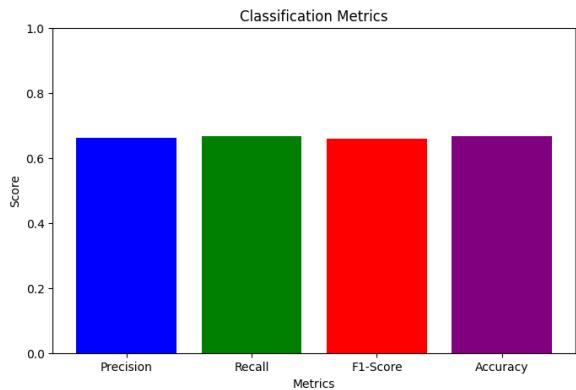


Figure 8: Heuristics-based Model Output

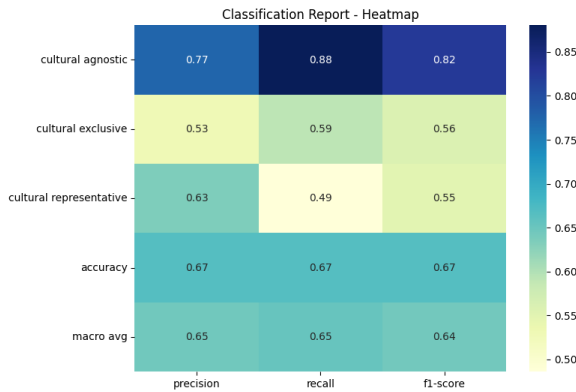


Figure 9: Heatmap of Model Behavior

A.3 Confusion Matrices

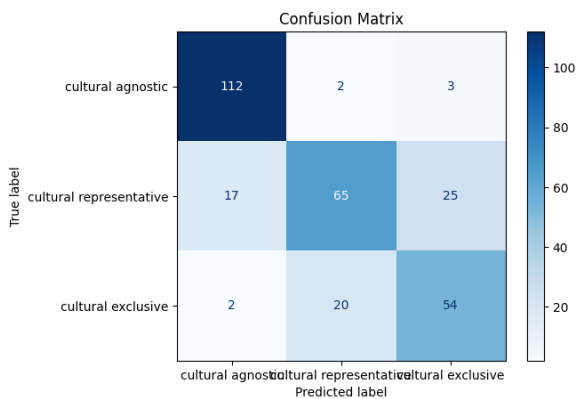


Figure 10: Confusion Matrix – LM-Based Model

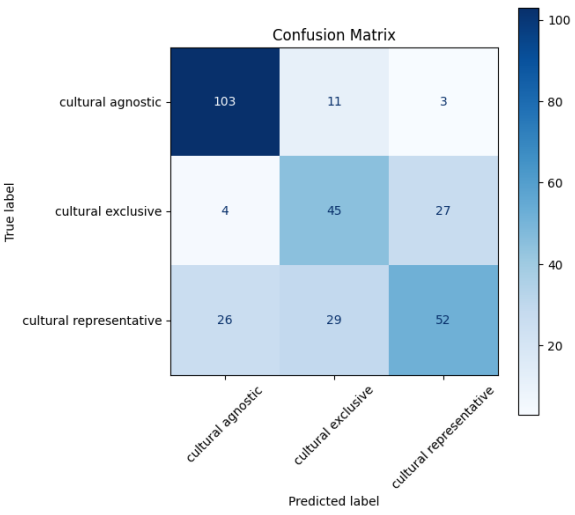


Figure 11: Confusion Matrix – Non LM-Based Model

A.4 Distribution of Predicted Classes



Figure 12: Distribution of Predicted Classes - LM-Based Model

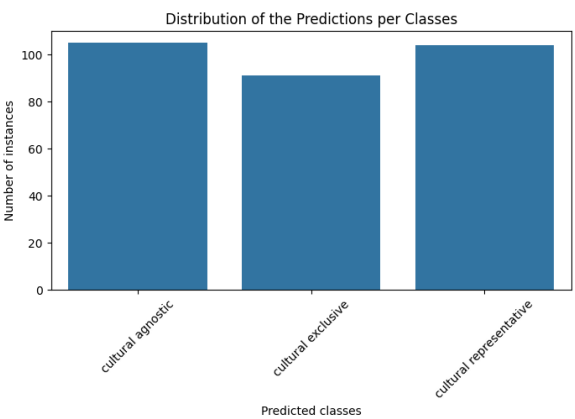


Figure 13: Distribution of Predicted Classes - Non LM-Based Model